

Learn Python

Student Edition

10 two-hour sessions · runnable examples · trap cheat sheets · self-check quizzes

An accelerated, self-study Python course
for a researcher with no prior coding experience.

Offline companion to the interactive website.
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Learn Python — Student Syllabus

Welcome. This is a fast, ~20-hour path (ten two-hour sessions) from "never coded" to "I can write a real Python program to wrangle my research data." It is re-ordered for you and front-loaded with the language quirks that trip people up.

How each 2-hour session runs: Concept (~25 min) → I code, you watch (~15 min) → **you code (~60 min, packed, split around a short break)** → Traps recap + quiz (~15 min). The practice block is the biggest on purpose — you learn by typing, and it holds enough tasks to fill the time. You will *type every example yourself*.

What you need: Python 3.11+, VS Code (or any editor), and this folder. Open the matching `slides/`, `examples/`, and `cheatsheets/` file for each session. (A full **PDF** of the course is downloadable from the website if you want to read offline.)

The 10 sessions (2 hours each)

#	Title	You'll be able to...	Files
1	Running Python, Variables & Types	Run code, use <code>int/float/str/bool/None</code> , do input/output with f-strings	<code>slides/session-01</code> · <code>examples/session-01</code>
2	The Dynamic-Typing Traps	Tell <code>==</code> from <code>is</code> , predict <code>True==1 / 0.1+0.2 / 5=="5"</code> , check types right	<code>slides/session-02</code> · <code>cheatsheets/traps-and-gotchas</code>
3	Control Flow: Conditionals & Loops	<code>if/elif/else</code> , chained comparisons, <code>for/while</code> , <code>break</code> , <code>enumerate/zip</code> , validation loops	<code>slides/session-03</code> · <code>examples/session-03</code>
4	Data Structures	Use <code>list/tuple/dict/set</code> , comprehensions, sorting, understand aliasing	<code>slides/session-04</code> · <code>examples/session-04</code>
5	Functions, Scope & Reuse	Write reusable functions, <code>*args/**kwargs</code> , type hints; dodge the mutable-default bug	<code>slides/session-05</code> · <code>examples/session-05</code>
6	Recursion & Recursive Thinking	Write functions that call themselves (base + recursive case), trace the call stack, recurse over nested data	<code>slides/session-06</code> · <code>examples/session-06</code>
7	Exceptions & Defensive Code	Validate messy input with <code>try/except</code> , raise errors, write a first test	<code>slides/session-07</code> · <code>examples/session-07</code>
8	Files, Libraries & Research Data	Read/write CSV survey data, use <code>statistics/datetime</code> , <code>pip install</code> , pandas teaser	<code>slides/session-08</code> · <code>examples/session-08</code>
9	Regular Expressions & Text Cleaning	Validate, extract, and clean real text with regex patterns & capture groups	<code>slides/session-09</code> · <code>examples/session-09</code>
10	Modules, OOP & the Pythonic Toolkit	Import modules; build a small class with <code>@property</code> ; use generators/ <code>map/ filter</code>	<code>slides/session-10</code> · <code>examples/session-10</code>
11	Capstone (optional)	Build a Gradebook & Survey Analyzer end-to-end	<code>assessments/capstone-project</code>

Session 2 is the most load-bearing session — if you deeply master one, make it that one.

Your standing toolkit (open these any time)

- `cheatsheets/traps-and-gotchas.md` — every quirk, with the wrong vs right way. *Keep this open.*
- `cheatsheets/quick-reference.md` — syntax you'll forget (slicing, f-strings, comprehensions).
- `cheatsheets/glossary.md` — plain-language definitions of every term.

How to study (specific to this material)

1. **Type, don't read.** Re-type every example in `examples/` ; change one thing and predict the result *before* you run it.
2. **Predict-then-run on traps.** For S2 especially, guess the output, then run. The surprise *is* the lesson.
3. **Get through the whole practice block.** It's sized for a fast learner — aim to finish every task, not just the first few.
4. **Keep a "bugs I hit" log.** When you get a traceback, write the last line + your fix. You'll build your own personalized cheat sheet.
5. **Connect to what you know.** Every session opens with how the concept maps to research methods/stats (see `connection-map.md`). Lean on those bridges.

What "done" looks like

You finish the capstone (S11) or, at minimum, complete every session's practice block and score the per-session quizzes in `assessments/` . The real test: you can open a messy CSV of your own data and write a short script that summarizes it without copying from anyone.

Scope (so you're not surprised)

This makes you a confident *programmer who handles research data*. It is **not** a full data-science course — pandas, plotting, and statistical modeling get a taste, not a deep dive. Those are the natural next step once these fundamentals are solid.

Session 1

Running Python, Variables & Types

Learn Python

Goal: run code, use the 5 core types, do input/output.

Why Python (for you)

- Free, readable, the lingua franca of research computing.
- Glue between your data, your stats, and your writing.
- Today: from zero to "I ran a program."

Mantra for the whole course: **readability wins.**

Two ways to run Python

1. **REPL** — type `python3`, get `>>>`, run one line at a time. Great for experiments.
2. **Script** — save `hello.py`, run `python3 hello.py`. Great for real work.

```
# hello.py
print("Hello, researcher")
```

Variables = labels on objects

```
n = 30          # an int
mean = 3.7      # a float
name = "Ada"    # a str
passed = True   # a bool
missing = None  # "no value"
```

`=` is an **action** ("stick this label on that object"), **not** a math equation. So `n = n + 1` is fine.

□ In stats you name quantities (`n`, `α`). Same idea — but the name can be re-pointed.

The 5 core types

Type	Example	Think
<code>int</code>	<code>30</code>	counts
<code>float</code>	<code>3.7</code>	measurements
<code>str</code>	<code>"Ada"</code>	text
<code>bool</code>	<code>True / False</code>	flags
<code>NoneType</code>	<code>None</code>	missing

`type(x)` tells you which.

Input & output

```
name = input("Your name: ")    # ⚠ ALWAYS a str
age = int(input("Your age: "))  # convert right away
print("Hi", name, "- age", age)
```

The #1 week-one trap: `input()` gives you text. `"5" + "3"` is `"53"`, not `8`.

f-strings (use these)

```
score = 87.456
print(f"{name} scored {score:.1f}")  # one decimal
print(f"{1234567:,}")                # 1,234,567
print(f"{0.873:.1%}")                # 87.3%
```

Cleaner than `+` concatenation and no type errors.

Type conversion (casting)

```
int("42")      # 42
float("3.14")   # 3.14
str(42)         # "42"
int(3.9)        # 3   (truncates, doesn't round!)
round(3.9)      # 4
```

`int(input(...))` = read text, convert to number, in one step.

Reading a traceback (don't panic)

```
Traceback (most recent call last):
  File "x.py", line 3, in <module>
    age = int(input("Age: "))
ValueError: invalid literal for int() with base 10: 'thirty'
```

Read the **LAST** line first. It names the problem: `ValueError`, and the bad value `'thirty'`.

Live demo & your turn

- Live: `greet.py`, then a "years to graduation" calculator. We'll break it once on purpose.
 - You: `examples/session-01/practice.md` — build a GPA-or-BMI style script.
-

Traps recap

- `input()` → **always a string**; convert with `int()` / `float()`.
- `print(a, b)` (comma → spaces) vs `print(a + b)` (must be same type).
- `int(3.9)` truncates to `3`; use `round()` to round.

Summary

You can run code, name values, convert types, format output, and read an error. **Next:** the dynamic-typing traps.

Session 1 — Practice (60 min)

Type each solution yourself. Predict output before running. Solutions at the bottom.

Task 1 — Warm-up REPL

In the REPL (`python3`), check the type of: `42` , `42.0` , `"42"` , `True` , `None` , `7/2` , `7//2` . What surprises you? (Hint: `7/2` .)

Task 2 — GPA reporter (`gpa.py`)

Ask for a name and a GPA (a decimal). Print: `"<name>'s GPA is <gpa to 2 decimals>, which is <87.5%> of a 4.0 scale."`

Task 3 — Survey age bucket (`age.py`)

Ask for an age (integer). Print the age, and the age in "months lived" ($\text{age} \times 12$). Then deliberately type `twenty` instead of a number and **read the traceback's last line**.

Task 4 — Stretch: the string trap

Without converting, what does `input("a: ") + input("b: ")` print if you type `2` then `3` ? Now fix it so it prints `5` .

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
a, b = 1, 2
a, b = b, a;          print(a, b)          # -> 2 1    (swap, no temp variable)
print(f"{a + b = }")   # -> a + b = 3    (self-documenting f-string)
print("CS50".lower(), " hi ".strip())      # -> cs50 hi
print("@" in "ana@uni.edu", len("data"))    # -> True 4    (membership + length)
```

Solutions

```
# Task 2 – gpa.py
name = input("Name: ")
gpa = float(input("GPA: "))
print(f"{name}'s GPA is {gpa:.2f}, which is {gpa/4:.1%} of a 4.0 scale.")

# Task 3 – age.py
age = int(input("Age: "))
print(f"You are {age} years old, about {age*12} months.")
# Typing "twenty" -> ValueError: invalid literal for int() with base 10: 'twenty'

# Task 4
# "2" + "3" -> "23" (string concatenation)
a = int(input("a: ")); b = int(input("b: "))
print(a + b)          # 5
```

Session 2

The Dynamic-Typing Traps

Predict every line before you run it.

Why this session matters

Python won't lock a variable to a type. That freedom creates **subtle traps**: mixing numbers, comparing across types, trusting floats.

Master these now → every later session is easier.

Method today: **predict** → **run** → **explain the surprise**.

Equality `==` vs Identity `is`

```
a = [1, 2]; b = [1, 2]
a == b      # True    same VALUE
a is b      # False   different OBJECTS
c = a
a is c      # True    same object (alias)
```

- `==` → "same value?" (almost always what you want)
- `is` → "same object?" (use only for `None`, `True`, `False`)

□ Same GPA = `==`. Same student = `is`.

Use `is` only for `None/True/False`

```
if x is None:      # ✓
if x == None:      # ⚠ works, not idiomatic
```

Wrinkle (never rely on it): CPython caches small ints (`-5...256`), so `256 is 256` → `True` but `257 is 257` may be `False`. For value, always `==`.

Booleans ARE integers

```
True == 1          # True
False == 0         # True
5 + True          # 6
sum([True, False, True]) # 2  <- counts the Trues
```

□ This is dummy coding (1/0) baked into the language. `sum(conditions)` = "how many are true?"

...but identity still differs

```
True == 1    # True    (value)
True is 1    # False   (different objects)
```

Compare with `isinstance(True, int)` → `True` (bool is a *subtype* of int).

Int vs float

```
3 == 3.0      # True    (compared by numeric value)
7 / 2         # 3.5     / ALWAYS returns a float
7 // 2        # 3       floor division
-7 // 2       # -4      floors toward -infinity
```

`4 / 2` is `2.0` (a float!), not `2`.

Float precision 🤖

```
0.1 + 0.2      # 0.30000000000000004
0.1 + 0.2 == 0.3 # False
```

Not a bug — decimals stored in **binary** can't all be exact.

```
import math
math.isclose(0.1 + 0.2, 0.3) # True  ✓
round(0.1 + 0.2, 2) == 0.3   # True  ✓
```

□ You already never `==` two measured scores. Same instinct, new cause.

Comparing across types

```
5 == "5"      # False   (different types, NO error)
5 == 5.0      # True    (numbers compare by value)
5 > "5"       # ❌ TypeError: '>' not supported
```

- `== / !=` across types → `False` (never crashes).
- `< / >` across incompatible types → **TypeError**.

❑ Computer can ask "same?" but can't *rank* text vs numbers — no shared scale.

Sequences compare element-by-element

```
[1, 2] == [1, 2]    # True
[1, 2] == (1, 2)    # False (list vs tuple = different types)
(1, 2) < (1, 3)     # True  (position by position)
"apple" < "banana"  # True  (char by char)
```

A list and a tuple with the same contents are **never equal**.

`type()` vs `instance()`

```
isinstance(x, int)          # ✔ Pythonic, respects inheritance
isinstance(x, (int, float)) # ✔ "any kind of number?"
type(x) is int              # exact type only
```

Gotcha: `isinstance(True, int)` is `True`. To exclude bools: `isinstance(x, int)` and not `isinstance(x, bool)`.

Truthiness

```
# Falsy: 0 0.0 "" [] {} set() None
# Truthy: everything else – including "0", "False", [0]
bool("0")    # True!
bool([0])    # True
```

Idiom: `if scores:` (not `if len(scores) > 0:`). But convert user text first — `"0"` is truthy.

Predict-then-run gauntlet

Cover the answers; commit out loud first.

```
True + True          # ?
3 == 3.0             # ?
0.1 + 0.2 == 0.3     # ?
5 == "5"             # ?
[1, 2] == (1, 2)      # ?
bool("0")            # ?
x=[1]; y=x; x.append(2); y  # ?
```

Your turn

examples/session-02/practice.md :

1. Explain each gauntlet line.
 2. Write `clean_score(value)` that accepts `5`, `5.0`, or `"5"` and returns a float, rejecting nonsense — safely.
-

Traps recap (take the cheat sheet)

- `==` value, `is` identity (only `None/True/False`).
- `True == 1`; `sum(bools)` counts.
- Never `==` floats → `math.isclose`.
- `5=="5" → False`; `5>"5" → TypeError`.
- list ≠ tuple even with same contents.
- `isinstance` over `type`; bools are ints.

Next: Control flow — conditionals & loops, now that you know what's *really* being compared.

Session 2 — Practice (60 min)

Part A — Explain the gauntlet (25 min)

For each line, write *why* (one sentence). Predict, then run to check.

```
True + True          # 2      - bools are ints; True is 1
3 == 3.0             # True   - numbers compare by value
0.1 + 0.2 == 0.3     # False  - binary float rounding
5 == "5"             # False  - different types, no error
5 > "5"              # □     - can't ORDER int vs str
[1,2] == (1,2)       # False  - list vs tuple are different types
bool("0")            # True   - non-empty string is truthy
x=[1]; y=x; x.append(2); y # [1,2] - y is an alias of x
```

Part B — `clean_score()` (35 min)

Write a function that safely turns a value into a float on a 0–100 scale:

```
def clean_score(value):
    """
    Accept 87, 87.0, or "87" and return 87.0 (a float).
    - Reject anything outside 0..100 with a clear message (return None).
    - Compare floats safely (no exact ==).
    """
```

Test it on: `87`, `87.0`, `"87"`, `"eighty"`, `120`, `True`. What does `True` do, and why? (Hint: `bool` is an `int`...)

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
x = int("257"); y = int("257")
print(x == y, x is y)          # -> True False    (equal value, different objects)
print(float("nan") == float("nan")) # -> False    (NaN equals nothing, not even itself)
```

Solution

```
import math

def clean_score(value):
    # Reject bools explicitly – they'd sneak through as ints (True == 1).
    if isinstance(value, bool):
        print(f"Rejected {value!r}: looks like a flag, not a score.")
        return None
    try:
        score = float(value)          # handles int, float, and numeric strings
    except (ValueError, TypeError):
        print(f"Rejected {value!r}: not a number.")
        return None
    if not 0 <= score <= 100:
        print(f"Rejected {value!r}: out of range 0-100.")
        return None
    return score

for v in [87, 87.0, "87", "eighty", 120, True]:
    print(v, "->", clean_score(v))
# 87->87.0, 87.0->87.0, "87"->87.0, "eighty"->None, 120->None, True->None
```

Key lesson: `float(True)` is `1.0`, so without the explicit bool check a flag would pass as a valid score. This is the `bool < int` trap in a real function.

Session 3

Control Flow: Conditionals & Loops

Make decisions, and repeat work — the two halves of control flow.

Part 1 — Conditionals

```
if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
else:
    grade = "F"
```

Indentation defines the block. Only the **first** true branch runs.

Comparison + chained comparisons

`==` `!=` `<` `<=` `>` `>=`

```
0 <= score <= 100      # chained – reads like math
```

□ No need for `score >= 0 and score <= 100` — chain it.

Logical operators (and the gotcha)

```
passed and submitted    # both
late or excused          # either
not flagged              # negate
```

Short-circuit: `a and b` skips `b` if `a` is falsy. And `and / or` return an **operand, not a bool**:

```
5 and 0      # 0
"" or "N/A"   # "N/A" ← default-value idiom
```

So write `if x:` — never `if x == True`.

Part 2 — Loops

```
for s in scores:           # each element directly
    print(s)

for i in range(5):         # 0,1,2,3,4
    print(i)

while not done:           # repeat until a condition flips
    ...
```

⚠ `range(1, 5)` → `1,2,3,4` — **stop is excluded** (off-by-one!).

break / continue

```
for x in data:
    if x is None:
        continue           # skip this one
    if x == "STOP":
        break              # leave the loop entirely
    process(x)
```

Stop juggling indices: enumerate & zip

```
for i, name in enumerate(names):    # index + value
    print(i, name)

for name, score in zip(names, scores): # two lists together
    print(name, score)
```

□ If you write `range(len(x))`, stop — use `enumerate` / `zip`.

The validation loop (you'll reuse this everywhere)

```
while True:
    raw = input("Score 0-100: ")
    if raw.isdigit() and 0 <= int(raw) <= 100:
        score = int(raw)
        break
    print("Try again.")
```

Your turn

`examples/session-03/practice.md`:

1. Grade-band classifier — test the boundaries (89.999 / 90 / 90.001).
 2. Average a roster and label each student PASS/FAIL with `zip`.
 3. A robust "ask until valid" loop.
-

Traps recap

- `if x == True` → just `if x:`; use `x is None` (not `== None`).
- `=` (assign) vs `==` (compare) — classic typo.
- `range(1, 5)` excludes 5; test your boundaries.
- Don't modify a list while looping it; prefer `enumerate / zip` over `range(len(...))`.

(More in the cheat sheet: `match / case`, the ternary, `for/else`.)

Summary

You can branch and repeat cleanly. **Next:** the containers you loop over — lists, dicts, sets.

Session 3 — Practice (60 min): Conditionals & Loops

Task 1 — Grade-band classifier

Write `letter_grade(score)` returning A/B/C/D/F (90/80/70/60 cutoffs), "Invalid" outside 0–100.

Test the boundaries: 90, 89.999, 0, 100, -5, 101.

Task 2 — Boolean logic

1. What is `5 and 0`? `""` or `"N/A"`? Why aren't they `True / False`?
2. Rewrite `if attended == True:` the Pythonic way.

Task 3 — Average + pass/fail (loops)

Given `names = ["Ana", "Ben", "Cara", "Dev"]` and `scores = [91, 58, 73, 64]`:

1. Compute the mean with a loop and a running total.
2. Use `zip` to print `"<name>: PASS"` (≥ 60) or `"<name>: FAIL"`.
3. Count the passes with `sum(s >= 60 for s in scores)`.

Task 4 — Validation loop

Write a real `while True:` prompt that keeps asking until the user types an integer 0–100.

Task 5 — Trap check

Why does this skip elements, and what's the fix?

```
xs = [1, 2, 3, 4]
for x in xs:
    if x % 2 == 0:
        xs.remove(x)
```

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
nums = [80, 92, 45]
print(all(n >= 60 for n in nums))    # -> False    (45 fails)
print(any(n >= 90 for n in nums))    # -> True     (92 passes)
print(92 in nums, 60 in nums)        # -> True False
```

Solutions

```
# 1
def letter_grade(score):
    if not 0 <= score <= 100: return "Invalid"
    for cutoff, g in [(90,"A"),(80,"B"),(70,"C"),(60,"D")]:
        if score >= cutoff: return g
    return "F"
# 90->A, 89.999->B, 0->F, 100->A, -5->Invalid, 101->Invalid

# 2
# 5 and 0 -> 0 ; "" or "N/A" -> "N/A" (and/or return an operand, not a bool)
result = "pass" if attended else "absent"      # and just `if attended:`

# 3
total = 0
for s in scores: total += s
print(total / len(scores))                    # 71.5
for name, score in zip(names, scores):
    print(f"{name}: {'PASS' if score >= 60 else 'FAIL'}")
print("passes:", sum(s >= 60 for s in scores)) # 3

# 4
while True:
    raw = input("Score 0-100: ")
    if raw.isdigit() and 0 <= int(raw) <= 100:
        print("Got", int(raw)); break
    print("Try again.")

# 5 Removing while iterating shifts indices, so elements get skipped.
xs = [x for x in xs if x % 2 != 0]            # build a new list instead -> [1, 3]
```

Session 4

Data Structures: list · tuple · dict · set

The containers that hold your data.

Four containers, four jobs

Type	Syntax	Mutable?	Use for
list	[1, 2, 3]	yes	ordered, changing collection
tuple	(1, 2)	no	fixed record / coordinates
dict	{"k": v}	yes	key → value lookup
set	{1, 2, 3}	yes	unique items

Lists & slicing

```
xs = [10, 20, 30, 40]
xs[0]      # 10      xs[-1]   # 40 (last)
xs[1:3]    # [20, 30] (stop excluded)
xs[:2]     # [10, 20]
xs[::-1]   # reversed
xs.append(50); xs.sort()      # mutate in place
```

⚠ `xs.sort()` returns **None** — it sorts in place. Use `sorted(xs)` for a new list.

Dicts = labeled records

```
student = {"name": "Ana", "gpa": 3.9}
student["name"]      # "Ana"
student.get("major", "N/A") # safe access with default
student["major"] = "Ed"    # add/update
for key, val in student.items(): ...
```

A list of dicts = a dataset ☐

```
roster = [  
    {"name": "Ana", "score": 91},  
    {"name": "Ben", "score": 58},  
]
```

Each dict = a **row/respondent**; each key = a **variable/column**. This is your tidy dataset until pandas shows up (Session 8).

Sets: unique, fast membership

```
answers = ["yes", "no", "yes", "maybe", "no"]  
set(answers)          # {'yes', 'no', 'maybe'} – dedup  
"yes" in set(answers)  # True, very fast
```

Great for "distinct responses" and "have I seen this ID?"

Comprehensions

```
[s["score"] for s in roster]          # list  
[s for s in roster if s["score"] >= 60] # with filter  
{s["name"]: s["score"] for s in roster} # dict  
[s["score"] // 10 for s in roster]    # set of score-decades
```

Read as: *expr, for each item, (optionally) if condition.*

Sorting with a key

```
sorted(roster, key=lambda s: s["score"])          # ascending  
sorted(roster, key=lambda s: s["score"], reverse=True) # descending
```

```
lambda s: s["score"] = "sort by the score field."
```

TRAP: aliasing (labels, not boxes)

```
a = [1, 2, 3]  
b = a          # SAME list  
a.append(4)  
b              # [1, 2, 3, 4] ☹️  
  
b = a.copy()   # ✔ independent copy
```

`[[0]*3]*3` makes 3 references to ONE row — use `[[0]*3 for _ in range(3)]`.

Your turn

`examples/session-04/practice.md`:

1. Build the roster (list of dicts); sort by score.
 2. `{name: score}` dict comprehension.
 3. Group students into pass/fail buckets.
 4. Demonstrate the aliasing trap and fix it.
-

Traps recap

- `=` aliases; use `.copy()` / `copy.deepcopy()`.
- `.sort()` returns None (in place); `sorted()` returns new.
- list $\neq$ tuple even with same contents (Session 2).
- `dict.get(key, default)` avoids `KeyError`.

Summary

You can store, look up, dedup, sort, and reshape data. **Next:** package logic into reusable functions.

Session 4 — Practice (60 min)

Start from:

```
roster = [
    {"name": "Ana", "score": 91}, {"name": "Ben", "score": 58},
    {"name": "Cara", "score": 73}, {"name": "Dev", "score": 64},
]
```

Task 1 — Rank

Print names sorted by score, highest first.

Task 2 — Map (dict comprehension)

Build `{name: score}` in one line.

Task 3 — Group

Build `{"pass": [...names...], "fail": [...names...]}` using a loop.

Task 4 — Dedup

From `["A", "B", "A", "C", "B"]`, get the distinct values and how many there are.

Task 5 — Aliasing

Show that `b = roster` then `roster.append({...})` also changes `b`. Then make `b` an independent copy so it doesn't. (Hint: nested dicts → `copy.deepcopy`.)

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
head, *tail = [10, 20, 30, 40]
print(head, tail)           # -> 10 [20, 30, 40]    (star-unpacking)
print({"a": 1} | {"b": 2})   # -> {'a': 1, 'b': 2}    (dict union, 3.9+)
print({1, 2, 3} & {2, 3, 4}) # -> {2, 3}          (set intersection)
print(list(zip(*[(1, 2), (3, 4)]))) # -> [(1, 3), (2, 4)] (transpose)
```

Session 5

Functions, Scope & Reusability

Define a rule once; apply it everywhere. (Reproducibility!)

Defining & calling

```
def class_average(scores):  
    """Return the mean of a list of scores."""  
    return sum(scores) / len(scores)  
  
class_average([91, 58, 73])    # 74.0
```

□ A function is a formula/coding-scheme: same input → same output.

return vs print

```
def avg(xs): return sum(xs) / len(xs)    # hands value back  
def show(xs): print(sum(xs) / len(xs))  # just displays  
  
x = avg([1,2,3])    # x = 2.0  
y = show([1,2,3])   # prints 2.0, but y is None!
```

`print` shows; `return` gives the value to the next step.

Parameters: positional, keyword, default

```
def grade(score, scale=100, passing=60):  
    ...  
grade(85)                # uses defaults  
grade(85, passing=50)    # keyword arg
```

⚠ Defaults must be **immutable** (numbers, strings, `None`) — never `[]` or `{}`.

args / *kwargs

```
def total(*args):          # any number of positionals -> tuple
    return sum(args)
total(1, 2, 3)             # 6

def tag(**kwargs):         # any number of keywords -> dict
    return kwargs
tag(name="Ana", gpa=3.9) # {'name': 'Ana', 'gpa': 3.9}

func(*my_list)             # unpack list into args
func(**my_dict)           # unpack dict into kwargs
```

TRAP: mutable default argument

```
def add_student(name, roster=[]):    # ✗
    roster.append(name)
    return roster

add_student("Ana")    # ['Ana']
add_student("Ben")    # ['Ana', 'Ben'] - the list PERSISTS!
```

The default `[]` is created **once**, at definition. Fix on next slide.

The fix: default to None

```
def add_student(name, roster=None):  # ✓
    if roster is None:
        roster = []
    roster.append(name)
    return roster
```

Rule: mutable default? Use `None` and create inside.

Scope (LEGB) & globals

Python looks up names: **Local** → **Enclosing** → **Global** → **Built-in**.

```
count = 0
def bump():
    count = count + 1    # □ UnboundLocalError
```

Assigning `count` makes it local. Avoid `global`; **return** a value and reassign instead.

Docstrings & type hints

```
def class_average(scores: list[float]) -> float:
    """Return the arithmetic mean of `scores`."""
    return sum(scores) / len(scores)
```

Type hints document intent. **They are NOT enforced at runtime** (`mypy` checks them).

Your turn

`examples/session-05/practice.md` :

1. A small grade-functions module (with docstrings + hints).
 2. Reproduce the mutable-default bug, then fix it.
 3. `summary(*scores)` using `*args`.
-

Traps recap

- Mutable default arg → use `None`.
- `print` ≠ `return` (forgot `return` → `None`).
- Assigning a global inside a function → `UnboundLocalError`.
- Type hints aren't enforced.

Summary

You can write reusable, documented, reproducible functions. **Next:** functions that call themselves — recursion.

Session 5 — Practice (60 min)

Task 1 — Grade-functions module

Write three functions with docstrings and type hints:

- `class_average(scores: list[float]) -> float`
- `letter_grade(score: float) -> str` (reuse Session 3)
- `pass_rate(scores: list[float], passing: float = 60) -> float` (fraction passing, 0–1)

Use bool-summing for `pass_rate` (recall `sum(s >= passing for s in scores)`).

Task 2 — Reproduce & fix the mutable-default bug

Write `add_note(text, notes=[])` that appends and returns. Call it three times and watch the list grow. Then fix it with the `None` pattern and prove each call starts fresh.

Task 3 — *args summary

Write `summary(*scores)` that returns a dict `{"n":..., "mean":..., "max":..., "min":...}`. Call it both as `summary(91, 58, 73)` and as `summary(*my_list)`.

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
def f(a, *, b):          # everything after * is keyword-only
    return a, b
print(f(1, b=2))          # -> (1, 2)
print(f(**{"a": 1, "b": 9})) # -> (1, 9)  (** unpacks a dict into arguments)
```

Solutions

```
def class_average(scores: list[float]) -> float:
    """Mean of scores."""
    return sum(scores) / len(scores)

def letter_grade(score: float) -> str:
    """A/B/C/D/F by 90/80/70/60 cutoffs."""
    for cutoff, letter in [(90, "A"), (80, "B"), (70, "C"), (60, "D")]:
        if score >= cutoff:
            return letter
    return "F"

def pass_rate(scores: list[float], passing: float = 60) -> float:
    """Fraction of scores >= passing (0..1)."""
    return sum(s >= passing for s in scores) / len(scores)

# Task 2
def add_note(text, notes=None):    # fixed version
    if notes is None:
        notes = []
    notes.append(text)
    return notes

# Task 3
def summary(*scores):
    return {"n": len(scores), "mean": sum(scores)/len(scores),
            "max": max(scores), "min": min(scores)}
print(summary(91, 58, 73))
print(summary(*[91, 58, 73]))
```

Session 6

Recursion & Recursive Thinking

A function that solves a problem by calling **itself** on a smaller piece.

The shape of every recursion

```
def countdown(n):
    if n <= 0:          # BASE CASE – when to stop
        print("liftoff!")
        return
    print(n)
    countdown(n - 1)    # RECURSIVE CASE – same problem, smaller input
```

Two parts, always:

- a **base case** that stops, and
 - a **recursive case** that moves *toward* the base case.
-

Trace the call stack

```
factorial(3)
= 3 * factorial(2)
= 3 * (2 * factorial(1))
= 3 * (2 * 1)          # base case returns 1
= 6
```

Each call waits on the one inside it. The calls stack up, then unwind.

□ Each pending call is a **stack frame** — that matters in a moment.

Recursion vs iteration

```
def factorial(n):
    if n <= 1:
        return 1
    return n * factorial(n - 1)    # ← must RETURN the call

def factorial_loop(n):
    total = 1
    for k in range(2, n + 1):
        total *= k
    return total
```

Same answer. For flat counting, the **loop** is usually clearer.

Where recursion shines: nested data

```
def deep_sum(obj):
    if isinstance(obj, (int, float)):
        return obj
    if isinstance(obj, dict):
        return sum(deep_sum(v) for v in obj.values())
    if isinstance(obj, (list, tuple)):
        return sum(deep_sum(x) for x in obj)
    return 0

deep_sum([1, [2, [3, 4]], {"a": 6}])    # 16
```

Nested JSON, folder trees, threaded replies — a single loop can't reach all the way down. Recursion can.

The trap: no base case

```
def runaway(n):
    return runaway(n + 1)    # never stops
```

```
RecursionError: maximum recursion depth exceeded
```

Python has **no tail-call optimization** — every call keeps its frame (default limit ≈ 1000). Deep recursion *will* hit the ceiling.

Your turn

examples/session-06/practice.md:

1. Recursive `rsum(n)` — name the base case first.
 2. `flatten([1, [2, [3, 4]], 5])` → one flat list.
 3. `depth(...)` — how deeply is a list nested?
-

Traps recap

- Every recursion needs a **reachable base case**, or it overflows the stack.
- **Return** the recursive call — forgetting to gives you a silent `None`.
- Recursion isn't free: each call costs a stack frame (no tail-call optimization).
- A plain **loop** is better for flat sequences and for very deep work.

Summary

You can solve problems that are defined in terms of themselves — especially nested data. **Next:** make your code survive messy real-world input — exceptions.

Session 6 — Practice (60 min): Recursion & Recursive Thinking

Task 1 — Recursive sum

Write `rsum(n)` that adds `1 + 2 + ... + n` **with recursion** (no loop). Name the base case out loud before you write it. Test `rsum(5)` and `rsum(0)`.

Task 2 — Recursion vs iteration

Write `reverse(s)` that reverses a string recursively. Then write the loop version. Which reads more clearly to you? Test `reverse("data")`.

Task 3 — Flatten nested data

Write `flatten(xs)` that turns a list-of-lists (nested to any depth) into one flat list: `flatten([1, [2, [3, 4]], 5])` → `[1, 2, 3, 4, 5]`. This is the move for nested JSON/exports.

Task 4 — How deep does it go?

Write `depth(xs)` returning how deeply a list is nested: `depth([1, [2, [3, [4]]]])` → `4`, `depth([1, 2, 3])` → `1`, `depth(5)` → `0`.

Task 5 — Trap check

1. Why does this raise `RecursionError`, and what's the fix? `python def f(n): return n + f(n - 1)`
2. This returns `None` instead of a number — why? `python def fact(n): if n <= 1: return 1 n * fact(n - 1)`
3. Name one case where a plain loop is the better choice over recursion.

Bonus — Pythonic idiom drill

One decorator makes exponential recursion instant by remembering past calls.

```
import functools

@functools.cache
def fib(n):
    return n if n < 2 else fib(n - 1) + fib(n - 2)
print(fib(35))
```

memoize: each n is computed once

-> 9227465 (try this WITHOUT @cache... then wait)

Solutions

```
# 1
def rsum(n):
    if n == 0:
        return 0
    return n + rsum(n - 1)
print(rsum(5), rsum(0))
```

base case

15 0

```
# 2
def reverse(s):
    if s == "":
        return ""
    return reverse(s[1:]) + s[0]
print(reverse("data"))
```

base case: empty string

all-but-first, reversed, then first

"atad"

loop version: "".join(reversed(s)) – usually clearer for flat strings

```
# 3
def flatten(xs):
    out = []
    for x in xs:
        if isinstance(x, list):
            out.extend(flatten(x))
        else:
            out.append(x)
    return out
print(flatten([1, [2, [3, 4]], 5]))
```

recurse into the sub-list

[1, 2, 3, 4, 5]

```
# 4
def depth(xs):
    if not isinstance(xs, list):
        return 0
    return 1 + max((depth(x) for x in xs), default=0)
print(depth([1, [2, [3, [4]]]]), depth([1, 2, 3]), depth(5))
```

a non-list has no nesting

4 1 0

```
# 5
# 1) No reachable base case -> the calls never stop -> stack overflows.
# Fix: add `if n == 0: return 0` (or n <= 0) at the top.
# 2) The recursive case computes n*fact(n-1) but never RETURNS it,
# so the function falls off the end and returns None. Add `return`.
# 3) A loop is better when the work is a simple flat sequence, or when the
# depth could exceed ~1000 (Python has no tail-call optimization, so deep
# recursion hits RecursionError where a loop would be fine).
```

Session 7

Exceptions & Defensive Code

Real research data is dirty. Make code that survives it.

try / except

```
try:
    n = int(value)
except ValueError:
    n = None          # handle the bad case
```

When code might fail at runtime, wrap it. The `except` catches the named error.

Common exception types

Exception	Happens when
<code>ValueError</code>	right type, bad value: <code>int("N/A")</code>
<code>TypeError</code>	wrong type: <code>5 > "5"</code>
<code>KeyError</code>	missing dict key: <code>d["nope"]</code>
<code>IndexError</code>	bad list index: <code>xs[99]</code>
<code>ZeroDivisionError</code>	<code>x / 0</code>
<code>FileNotFoundError</code>	<code>open("missing.csv")</code>

try / except / else / finally

```
try:
    n = int(value)
except ValueError:
    print("not a number")
else:
    print("ok:", n)          # only if NO exception
finally:
    print("always runs")    # cleanup
```

Raise your own

```
def clean_likert(n):
    if not 1 <= n <= 5:
        raise ValueError(f"{n} not in 1-5")
    return n
```

`raise` throws an exception on purpose — caller decides how to handle it.

EAFP vs LBYL

```
# LBYL – "look before you leap"
if value.isdigit():
    n = int(value)

# EAFP – "easier to ask forgiveness" (Pythonic)
try:
    n = int(value)
except ValueError:
    n = None
```

Both valid. EAFP shines when "checking first" is hard or racy.

assert (developer check, not validation)

```
assert len(scores) > 0, "scores must not be empty"
```

For *your* sanity checks while developing. Can be disabled (`python -O`), so **never** use `assert` to validate untrusted input — use `raise`.

A first test with pytest

```
# clean.py
def clean_likert(n):
    if not 1 <= n <= 5:
        raise ValueError("1-5 only")
    return n

# test_clean.py
import pytest
from clean import clean_likert

def test_valid():
    assert clean_likert(3) == 3
def test_invalid():
    with pytest.raises(ValueError):
        clean_likert(9)
```

Run: `pytest`

Your turn

`examples/session-07/practice.md`:

1. `safe_int(value)` returning int or None.
 2. Clean a dirty survey list, collecting good values + a rejection log.
 3. Write one `pytest` test.
-

Traps recap

- **Never** bare `except:` — name the exception.
- Don't catch too broadly or swallow errors silently.
- `assert` ≠ input validation (use `raise`).
- Catch the *specific* error you expect.

Summary

You can validate messy input and fail loudly when you should. **Next:** read that messy data from real files.

Session 7 — Practice (60 min)

Task 1 — `safe_int`

Write `safe_int(value)` returning `int(value)` or `None` on failure. Test on `"42"`, `"N/A"`, `""`, `None`, `3.0`.

Task 2 — Clean a survey column

Given `raw = ["5", "3", "N/A", "7", "", "1", "two", "4"]`, produce:

- `clean` — list of valid Likert ints (1–5), and
- `rejected` — list of `(value, reason)` pairs. Use a `clean_likert(n)` that **raises** `ValueError` for out-of-range or non-ints.

Task 3 — Write a test

Put `clean_likert` in `clean.py` and write `test_clean.py` with pytest: one passing case and one `pytest.raises(ValueError)` case. Run `pytest`.

Task 4 — Discuss

Why is `except:` (bare) dangerous? Give one error it would hide that you'd rather see.

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
class LikertError(ValueError):          # your own exception type
    pass
print(issubclass(LikertError, ValueError)) # -> True (so `except ValueError` still
catches it)

try:
    assert 1 == 2, "values differ"      # assert: cheap internal sanity check
except AssertionError as e:
    print(e)                           # -> values differ
```

Solutions

```
def safe_int(value):
    try:
        return int(value)
    except (ValueError, TypeError):
        return None

def clean_likert(n):
    if isinstance(n, bool) or not isinstance(n, int):
        raise ValueError(f"{n!r} not an int")
    if not 1 <= n <= 5:
        raise ValueError(f"{n} outside 1-5")
    return n

raw = ["5", "3", "N/A", "7", "", "1", "two", "4"]
clean, rejected = [], []
for r in raw:
    try:
        clean.append(clean_likert(safe_int(r)))
    except ValueError as e:
        rejected.append((r, str(e)))
print(clean)      # [5, 3, 1, 4]
print(rejected)   # [('N/A', ...), ('7', ...), ('', ...), ('two', ...)]
```

```
# test_clean.py
import pytest
from clean import clean_likert
def test_ok():    assert clean_likert(3) == 3
def test_bad():
    with pytest.raises(ValueError):
        clean_likert(9)
```

Task 4: a bare `except:` also catches `KeyboardInterrupt` (Ctrl+C) and `NameError` from your own typos — so a misspelled variable would be silently swallowed instead of showing you the bug. Always catch the specific exception you expect.

Session 8

Files, Libraries & Research Data

The session your actual data shows up.

Opening files with `with`

```
with open("notes.txt") as f:
    text = f.read()
# file auto-closes here, even if the code crashes
```

`with` = a context manager: sets up and tears down the resource for you. Always prefer it to a bare `open()` / `close()`.

File modes (mind the trap)

Mode	Meaning
"r"	read (default)
"w"	write — truncates the file to empty first!
"a"	append
"r+"	read + write

⚠ Open the wrong file with `"w"` → its contents are gone.

Reading text

```
with open("notes.txt") as f:
    whole = f.read()          # one big string
    # or
    for line in f:             # line by line (memory-friendly)
        print(line.rstrip())
```

⚠ A file object is exhausted after one pass — re-open to read again.

CSV in, as dicts □

```
import csv
with open("students.csv", newline="") as f:
    for row in csv.DictReader(f):
        print(row["name"], row["score"])    # row is a dict keyed by header
```

`csv.DictReader` turns each row into a dict — your "list of dicts" dataset from Session 4. (`newline=""` avoids blank rows on Windows.)

CSV out

```
with open("summary.csv", "w", newline="") as f:
    w = csv.DictWriter(f, fieldnames=["name", "score"])
    w.writeheader()
    w.writerow({"name": "Ana", "score": 91})
```

Libraries a researcher reaches for

```
import statistics
statistics.mean(xs); statistics.median(xs); statistics.stdev(xs)

import random
random.choice(xs); random.randint(1, 6); random.shuffle(xs)

from datetime import date
date.today()

from pathlib import Path
Path("students.csv").exists()
```

`pip install <package>` for third-party libs.

The pandas teaser (your next course)

```
import pandas as pd
df = pd.read_csv("students.csv")
df["score"].describe()    # count, mean, std, min, quartiles, max
df.groupby("major")["score"].mean()
```

Everything you did by hand today — in three lines. We learned the fundamentals *underneath* it first.

Your turn

`examples/session-08/practice.md` (uses `survey.csv`):

1. Read `students.csv`; print class mean with `statistics.mean`.
 2. Compute per-item survey means; write `survey_summary.csv`.
-

Traps recap

- `"w"` silently overwrites — be sure of the filename.
- `csv` module → open with `newline=""`.
- Files exhaust after one read; re-open to re-read.
- Specify `encoding="utf-8"` for non-ASCII text.

Summary

You can load, summarize, and write real research data. **Next:** clean and validate text with regular expressions.

Session 8 — Practice (60 min)

Files provided: `students.csv`, `survey.csv`.

Task 1 — Read & summarize students

Read `students.csv` with `csv.DictReader`. Remember the values are **strings** — convert `score` to `int`. Print the class mean and median with the `statistics` module.

Task 2 — Mean by major

Build `{major: mean_score}`. (Hint: `dict.setdefault(key, []).append(...)`.)

Task 3 — Clean & summarize the survey

`survey.csv` has "N/A" and blanks in numeric columns. For each `q*` item, compute the mean of the **valid** values only, and how many were valid. Write `survey_summary.csv` with columns `item, mean, n_valid`.

Task 4 — pandas teaser (optional)

If `pandas` is installed: `pd.read_csv("students.csv")["score"].describe()`. Compare the mean to your hand-computed one.

Trap check

What happens if you accidentally open `students.csv` with mode `"w"` before reading it?

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
import json
s = json.dumps({"n": 3, "ok": True})
print(s)                                # -> {"n": 3, "ok": true}   (Python True -> JSON true)
print(json.loads(s)["ok"])              # -> True           (and back to a Python
bool)
```

Solutions

See `demo.py` in this folder — it implements Tasks 1–3 exactly. Key lines:

```
scores = [int(s["score"]) for s in students]    # convert strings!
statistics.mean(scores)                        # 75.5

by_major = {}
for s in students:
    by_major.setdefault(s["major"], []).append(int(s["score"]))

def to_int(x):
    try: return int(x)
    except (ValueError, TypeError): return None    # handles N/A and ""
```

Trap: opening with `"w"` **truncates the file to empty immediately** — your data is gone before you ever read it. Use `"r"` (the default) to read.

Session 9

Regular Expressions & Text Cleaning

Find, validate, extract, and clean messy text — the researcher's power tool.

Why a researcher cares

- Validate IDs, emails, dates before they pollute your data.
- Extract structured bits from free text (codes, names, numbers).
- Clean and normalize open-ended survey responses.
- A first pass at **qualitative coding** (find every response matching a pattern).

□ Like search-and-filter over a corpus — but it matches *form*, not meaning.

Always use raw strings

```
import re
re.search(r"\d+", "id 42")    # r"..." = raw string
```

Without `r"..."`, Python eats the backslashes (`\d` → error/garbage). **Rule:** every regex pattern is a raw string.

The survival tokens

Token	Matches
<code>.</code>	any char (except newline)
<code>\d \w \s</code>	digit / word-char / whitespace
<code>\D \W \S</code>	the negations
<code>+ * ?</code>	1+, 0+, 0-or-1
<code>{m} {m, n}</code>	exactly m / between m and n
<code>^ \$</code>	start / end of string
<code>[abc] [^abc]</code>	any in set / none in set
<code>(...)</code>	capture group
<code>a\b</code>	a or b

The `.` trap

```
re.search(r".", "a.b").group()    # 'a' - ANY char, not a dot!
re.search(r"\.", "a.b").group()   # '.' - escape it for a literal dot
```

Escape the specials when you mean them literally: `\. \^ \$ \* \+ \? \(\ \) \[ \] \{ \} \|`

The four functions you need

```
re.search(pattern, s)      # first match ANYWHERE -> match or None
re.fullmatch(pattern, s)   # the WHOLE string must match -> validation
re.findall(pattern, s)     # list of ALL matches
re.sub(pattern, repl, s)   # replace matches -> cleaning
```

`re.IGNORECASE` flag for case-insensitive: `re.search(p, s, re.IGNORECASE)`.

Validate (fullmatch anchors both ends)

```
def valid_university_email(addr):
    return re.fullmatch(r"\w+@\w+\.edu", addr) is not None

valid_university_email("ana@university.edu")    # True
valid_university_email("ana@gmail.com")         # False
valid_university_email("ana@x.edu")             # False
```

Extract with capture groups

```
m = re.search(r"([A-Z]{2})(\d{4})", "Course ED1234 meets Tue")
m.group(0)    # "ED1234"  whole match
m.group(1)    # "ED"      dept
m.group(2)    # "1234"    number
```

`m` is `None` if nothing matched — check before `.group()`.

Clean & mine free text

```
re.sub(r"\s+", " ", messy).strip()    # collapse whitespace
re.findall(r"#(\w+)", "love #python #stats") # ['python', 'stats']

from collections import Counter
Counter(re.findall(r"#(\w+)", corpus))   # theme frequencies
```

Reformat with groups: `re.sub(r"^(.+),\s*(.+)$", r"\2 \1", "Curie, Marie")` → `"Marie Curie"`.

When NOT to use regex

```
"a,b,c".split(",")      # simple split – no regex needed
"  hi  ".strip()        # trim – no regex needed
text.replace("X", "Y")  # fixed substring – no regex needed
```

Regex shines for *variable* patterns. For fixed strings, plain methods read better.

Your turn

`examples/session-09/practice.md`:

1. Email validator.
 2. Extract dept+number.
 3. Collapse whitespace.
 2. Count hashtags across responses.
 5. Flip `"Last, First"`.
 6. One case to use `.split()` instead.
-

Traps recap

- `.` matches **any** char — use `\.` for a literal dot.
- Forgetting `r"..."` breaks your backslashes.
- `re.search` returns `None` on no match — guard before `.group()`.
- Don't use regex where a string method is clearer.

Summary

You can validate, extract, and clean real-world text. **Next:** organize code into modules and classes.

Session 9 — Practice (60 min): Regular Expressions

Always use raw strings `r"..."`. Predict each result before running.

Task 1 — Validate

Write `valid_university_email(addr)` returning `True` only for `something@something.edu`. Test: `"ana@university.edu"`, `"ana@gmail.com"`, `"a@b.edu.evil.com"`.

Task 2 — Extract with groups

From `"Course ED1234 meets Tue"`, pull the department (`ED`) and number (`1234`) using one regex with two capture groups.

Task 3 — Clean

Collapse all runs of whitespace in `" too much\t space "` to single spaces and trim.

Task 4 — Mine free text

From a list of open-ended responses, count how often each `#hashtag` appears (use `re.findall(r"#(\w+)", text)` and `collections.Counter`).

Task 5 — Reformat

Turn `"Curie, Marie"` into `"Marie Curie"` with a single regex + groups.

Task 6 — Judgment

Give one task where a plain string method (`.split()`, `.strip()`, `.replace()`) is the better, clearer choice than a regex.

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```
import re
m = re.search(r"(?P<year>\d{4})", "class of 2026")
print(m.group("year"), m.groupdict()) # -> 2026 {'year': '2026'} (named groups)
print(re.split(r"\s*,\s*", "a, b ,c")) # -> ['a', 'b', 'c'] (split on commas + spaces)
```

Solutions

See `demo.py` in this folder — it implements all six. Key lines:

```
re.fullmatch(r"\w+@\w+\.\edu", addr) is not None      # 1 (fullmatch anchors both ends)
m = re.search(r"([A-Z]{2})(\d{4})", s); m.group(1), m.group(2)  # 2
re.sub(r"\s+", " ", messy).strip()                    # 3
from collections import Counter; Counter(re.findall(r"#(\w+)", text))  # 4
m = re.search(r"^(.+),\s*(.+)$", s); f"{m.group(2)} {m.group(1)}"  # 5
```

Task 6: splitting `"a,b,c"` on commas is just `"a,b,c".split(",")` — no regex needed. Reach for regex only when the pattern is genuinely variable (digits, optional parts, anchors).

Trap reminder: `.` matches **any** character — use `\.` for a literal dot, and never forget the `r"..."` prefix or your backslashes become Python escape sequences.

Session 10

Modules, OOP & the Pythonic Toolkit

Organize code into reusable pieces — and finish with the moves experts use.

Modules: split your code into files

```
# grades.py
def letter_grade(score): ...
def class_average(scores): ...

# analysis.py
from grades import letter_grade, class_average
```

A `.py` file is a **module**. `import` reuses its functions elsewhere → no copy-paste.

The `__main__` guard

```
# grades.py
if __name__ == "__main__":
    print(letter_grade(85))    # runs ONLY when you execute grades.py directly
```

On `import grades`, `__name__` is `"grades"`, so the block is skipped. One file can be both a runnable script *and* an importable library.

OOP: model a domain entity

```
class Student:
    def __init__(self, name, gpa):    # constructor
        self.name = name
        self.gpa = gpa
    def __str__(self):                # how it prints
        return f"{self.name} ({self.gpa})"

ana = Student("Ana", 3.9)
print(ana)    # Ana (3.9)
```

□ A class is an *operational definition*: the attributes + behaviors that "count" as a Student. `self` = "this particular student."

@property: validate on assignment

```
class Student:
    ...
    @property
    def gpa(self):
        return self._gpa
    @gpa.setter
    def gpa(self, value):
        if not 0 <= value <= 4:
            raise ValueError("gpa must be 0-4")
        self._gpa = value
```

`ana.gpa = 5.0` now raises — the object defends its own integrity.

Inheritance

```
class GradStudent(Student):
    def __init__(self, name, gpa, advisor):
        super().__init__(name, gpa)    # reuse parent setup
        self.advisor = advisor
    def __str__(self):
        return super().__str__() + f" - {self.advisor}"
```

`GradStudent` is a `Student` plus extra. `super()` calls the parent.

The Pythonic toolkit (recap tour)

```
[s.name for s in roster if s.gpa >= 2.0]    # comprehension (from S4)
list(map(lambda s: s.name.upper(), roster)) # map: apply to all
list(filter(lambda s: s.gpa < 2.0, roster)) # filter: keep matches
for i, s in enumerate(roster): ...          # index + item
for a, b in zip(names, scores): ...         # parallel
```

Generators & walrus

```
def gpas(students):
    for s in students:
        yield s.gpa    # one value at a time – low memory on big data

g = gpas(roster)
list(g)    # [3.9, 1.8, ...]
list(g)    # [] ← a generator is exhausted after one pass

if (n := len(roster)) > 30:    # walrus := : assign + test in one step
    print(f"{n} students")
```

Your turn

`examples/session-10/practice.md`:

1. Import from `grades.py`.
 2. Build the validating `Student` class.
 3. Add `GradStudent` with `super()`.
 4. Comprehension + `map` + `filter` + a generator.
-

Traps recap

- `self` is just "this instance" — not magic.
- A generator iterates **once**, then it's empty.
- The `__main__` guard keeps imported modules from running their demo code.
- Don't reach for a class when a function or dict will do.

Summary

You can structure code into modules and classes and write idiomatic Python. **Next (optional):** the capstone — put it all together.

Session 10 — Practice (60 min): Modules, OOP & Pythonic

Files in this folder: `grades.py` (a module you import), `demo.py` (worked example).

Task 1 — Use a module

From a new file, `from grades import letter_grade, class_average` and call both. Why does the `if __name__ == "__main__":` block in `grades.py` NOT run when you import it?

Task 2 — A class with a validating property

Build `Student(name, gpa)` with:

- `__str__` → `"Ana: 3.9 (Good)"`,
- `standing()` → `"Good"` if `gpa ≥ 2.0` else `"Probation"`,
- a `@property` setter for `gpa` that raises `ValueError` outside 0–4. Prove the setter rejects `5.0`.

Task 3 — Inheritance

Add `GradStudent(Student)` that also stores an `advisor` and uses `super().__init__(...)`. Override `__str__` to append the advisor.

Task 4 — The Pythonic toolkit

Given a roster of `Student` s:

1. names in good standing (list comprehension),
2. uppercase names (`map`),
3. at-risk students (`filter`),
4. mean gpa via a **generator** that `yield`s each gpa — then show the generator is empty on a second pass.

Bonus — Pythonic idiom drill

Cover the `# ->` answers, predict each line, then run.

```

from dataclasses import dataclass

@dataclass                                # auto __init__, __repr__, __eq__
class Point:
    x: int
    y: int
print(Point(1, 2), Point(1, 2) == Point(1, 2))    # -> Point(x=1, y=2) True

def head(v):
    match v:                                # structural pattern matching (3.10+)
        case [first, *_]: return first
        case _: return None
print(head([9, 8]), head(5))                  # -> 9 None

```

Solutions

See `demo.py` — it implements Tasks 2–4. Key points:

```

@property
def gpa(self): return self._gpa
@gpa.setter
def gpa(self, v):
    if not 0 <= v <= 4: raise ValueError(...)
    self._gpa = v

class GradStudent(Student):
    def __init__(self, name, gpa, advisor):
        super().__init__(name, gpa)
        self.advisor = advisor

[s.name for s in roster if s.gpa >= 2.0]          # comprehension
list(map(lambda s: s.name.upper(), roster))      # map
[s.name for s in filter(lambda s: s.gpa < 2.0, roster)] # filter
def gpas(rs):
    for s in rs: yield s.gpa                      # generator (exhausts after one pass)

```

Task 1: the `__name__` guard is only `"__main__"` when the file is **run directly**; on `import` its `__name__` is `"grades"`, so the demo block is skipped — that's how a file can be both a runnable script and an importable module.

Setup & Tools

Everything you need to run Python on your own machine — plus the tools (and AI habits) that make learning this material faster.

Fastest start: no install at all

Every example on this site runs **in your browser** — just press **Run**. Nothing to install, great for the first sessions. When you want to work on *your own files*, install Python locally with the steps below.

Run the course as Jupyter notebooks

Every session is also a **Jupyter notebook** — the same examples and practice as a runnable, cell-by-cell document. Three ways to use them; pick whatever fits:

- **In your browser, no install** — open the live JupyterLite workspace: <https://yangnei.github.io/learn-python/jupyter/lab/index.html>. It's real Jupyter running entirely in your browser; your edits are saved locally in the browser, nothing is uploaded. Each session page also has a **Run in browser** button that opens that session's notebook directly.
- **Google Colab** — each session page has an **Open in Colab** button (free, runs in Google's cloud, needs a Google account). Handy when you want to save to Drive or do heavier data work.
- **Local Jupyter** — once Python is installed (below): `pip install jupyterlab`, then run `jupyter lab` and open the `.ipynb` you saved with a session page's **Download .ipynb** button.

How to drive a notebook: click a cell and press **Shift + Enter** to run it and move to the next. Predict the output before you run — the surprise is the lesson — then edit a cell and re-run to experiment.

Install Python (pick your OS)

Windows

- Official installer: <https://www.python.org/downloads/windows/> — on the first screen, **check "Add python.exe to PATH"**, then *Install Now*.
- (Alternative) Microsoft Store → search **Python 3.12**.
- Check it worked — open **PowerShell** and run: `python --version`
- Official guide: <https://docs.python.org/3/using/windows.html>

macOS

- Official installer: <https://www.python.org/downloads/macos/>
- (Alternative) **Homebrew**: `brew install python`

- Check it worked — open **Terminal** and run: `python3 --version`
- Official guide: <https://docs.python.org/3/using/mac.html>

Linux

- Usually already installed. If not: `sudo apt install python3 python3-pip` (Debian/Ubuntu) or `sudo dnf install python3` (Fedora).
- Check it worked: `python3 --version`
- Official guide: <https://docs.python.org/3/using/unix.html>

An editor: install **VS Code** (<https://code.visualstudio.com/>) and its **Python extension** (by Microsoft). Step-by-step: <https://code.visualstudio.com/docs/python/python-tutorial>. Prefer something lighter? **Thonny** (<https://thonny.org/>) is a beginner IDE with a built-in variable viewer and debugger.

Run a script: save your code as `name.py`, then in a terminal run `python name.py` (Windows) or `python3 name.py` (macOS/Linux).

Install a package (e.g. for the Session 8 pandas teaser): `pip install pandas` (or `pip3`).

Visualize what your code does — Python Tutor

<https://pythontutor.com/> — paste code and **step through it line by line**, watching variables, the call stack, and which names point at which objects. It's the single best tool for the things this course keeps warning you about:

- **aliasing** — see `b = a` make both names point at the *same* list (Session 4),
- the **mutable-default-argument** bug (Session 5),
- the **recursion call stack** building up and unwinding (Session 6).

Whenever you think *"wait, why did that happen?"* — drop the snippet into Python Tutor.

Use AI to learn (without it doing the work for you)

An AI assistant (Claude, etc.) is a huge accelerator *if you ask it to explain rather than to write*. Type every solution yourself. Prompts that fit this course:

- **"Summarize this doc page for a beginner, just the 5 things I need: \<paste text or URL>"** — turn a dense reference page into something usable.
- **"Explain this error and its most likely cause, and point me to the line: \<paste the traceback>"**
- **"Predict what this prints, then explain why: \<code>"** — then run it and compare. That predict-then-run habit is the method this course is built on.
- **"Review my function for the traps in this course — identity vs value, aliasing, mutable defaults, off-by-one — but don't rewrite it for me."**

- "Give me 3 harder practice tasks like this one, with the solutions hidden."

Rule of thumb: **ask it to explain, not to do**. If it hands you code you can't read line by line, ask it to slow down and walk you through it.

Other handy tools

- **regex101.com** (<https://regex101.com/>) — build and *explain* regular expressions interactively, with a live match view (Session 9). Set the flavor to **Python**.
- **Official docs** — the [Python Tutorial](#) and the [Library Reference](#). Bookmark both.
- **Google Colab** (<https://colab.research.google.com/>) — run Python notebooks in the browser, no install; a natural next step once you start doing real data work.
- This site's [Traps & Gotchas](#) sheet — keep it open the whole course.

Python Traps & Gotchas — The Master Cheat Sheet

The quirks that bite beginners (and plenty of experts). Every behavior below was run and verified on CPython 3.11+. Read the **Surprise** column, cover the **Why/Fix**, and predict. Keep this open for the whole course.

1 — Equality `==` vs Identity `is` (Session 2)

Expression	Result	
<code>a = [1,2]; b = [1,2]; a == b</code>	<code>True</code>	same value
<code>a = [1,2]; b = [1,2]; a is b</code>	<code>False</code>	different objects in memory
<code>a = [1,2]; b = a; a is b</code>	<code>True</code>	<code>b</code> is the <i>same</i> object (an alias)

- `==` asks "same value?" — this is what you almost always want.
- `is` asks "same object?" — use it **only** for `None`, `True`, `False`: `python if x is None: # ✓`
`correct if x == None: # ⚠ works, but not idiomatic – use `is``
- **Identity wrinkle (do NOT rely on this):** CPython caches small integers (–5 to 256) and interns some strings, so `a = 256; b = 256; a is b` is `True` but at `257` it may be `False`. This is an implementation detail. **For value, always use `==`.**

□ *Bridge:* two students with the same GPA are `==`; the same student is `is`.

2 — Booleans ARE integers (Session 2)

```
True == 1          # True    (bool is a subclass of int)
False == 0         # True
5 + True          # 6        yes, you can do arithmetic on bools
sum([True, False, True]) # 2 ← counts the Trues
isinstance(True, int) # True
type(True) is int   # False   (its type is bool, a subtype)
```

- **Fix / use it well:** `sum(conditions)` is the Pythonic way to *count* how many are true.
- **Trap:** `True is 1` is `False` (different objects) even though `True == 1`. Compare with `==`.

□ *Bridge:* this is dummy coding (1/0) built into the language — summing flags counts cases.

3 — Float precision (Session 2)

```
0.1 + 0.2          # 0.30000000000000004
0.1 + 0.2 == 0.3    # False 🤖
```

- **Why:** decimals are stored in binary; most can't be represented exactly. Not a Python bug — every language does this.
- **Fix:** `python import math math.isclose(0.1 + 0.2, 0.3) # True` ← compare floats with a tolerance `round(0.1 + 0.2, 2) == 0.3 # True` ← or round for display/compare from decimal `import Decimal # exact decimal arithmetic when you truly need it (money, grades) Decimal("0.1") + Decimal("0.2") # Decimal('0.3')`
- **Rule:** never test computed floats with `==`. Use `math.isclose` (or `round`).

□ *Bridge:* you already never test two measured scores for exact equality — same instinct, new cause (binary storage, not measurement error).

4 — Comparing across types (Session 2)

```
5 == "5"          # False   (different types → not equal, but NO error)
5 == 5.0           # True    (int vs float compare by numeric value)
5 > "5"            # ❌ TypeError: '>' not supported between 'int' and 'str'
```

- **Equality** (`== / !=`) across unrelated types → just `False`, never crashes.
- **Ordering** (`<, >, <=, >=`) across incompatible types → `TypeError`.
- **Fix:** convert first. `int("5") == 5 → True`. Guard with `isinstance` before ordering.

□ *Bridge:* the computer can equate "are these the same?" across types, but can't *rank* text against numbers — it has no shared scale.

5 — Sequences compare element-by-element (Session 2 / 5)

```
[1, 2] == [1, 2]    # True
[1, 2] == (1, 2)     # False   ← list vs tuple: different types, never equal
(1, 2) < (1, 3)       # True    ← compares position by position (lexicographic)
"apple" < "banana"    # True    ← strings compare char by char (alphabetical-ish)
[1, 2] < [1, 2, 3]    # True    ← prefix is "less than"
```

- **Trap:** a `list` and a `tuple` with identical contents are **never** `==` — type matters.
- Lists/tuples/strings compare left-to-right; the first differing element decides.

6 — Truthiness (what counts as False) (Session 2 / 3)

```
# These are all "falsy":
bool(0) bool(0.0) bool("") bool([]) bool({}) bool(set()) bool(None) # all False
bool("0") # True! ← a non-empty string is truthy, even "0"
bool([0]) # True ← a non-empty list is truthy, even if it holds a 0
```

- **Fix / idiom:** check emptiness directly — `if scores: not if len(scores) > 0;` ; `if name: not if name != ""` ;
- **Trap:** `"0"` (string) and `"False"` (string) are **truthy**. Convert user input before testing.

7 — `and` / `or` return an operand, not a bool (Session 3)

```
5 and 0 # 0 (and → first falsy, or last value)
0 or "hi" # "hi" (or → first truthy, or last value)
"" or "N/A" # "N/A" ← handy default-value idiom
```

- **Use it well:** `name = user_input or "Anonymous"` supplies a default.
- **Trap:** don't write `if x == True`. Just `if x:`. And `and` / `or` short-circuit — the right side may never run, so don't hide side effects there.

8 — Mutable default arguments (Session 6) — the famous one

```
def add_student(name, roster=[]): # ✗ DANGER
    roster.append(name)
    return roster

add_student("Ana") # ['Ana']
add_student("Ben") # ['Ana', 'Ben'] ☹️ the default list PERSISTS across calls
```

- **Why:** the default `[]` is created **once**, when the function is *defined*, and reused every call.
- **Fix:** `python def add_student(name, roster=None): # ✓ if roster is None: roster = []`
`roster.append(name) return roster`
- **Rule:** never use a mutable default (`[]`, `{}`, `set()`). Use `None` and create inside.

9 — Aliasing: variables are labels, not boxes (Session 5)

```
a = [1, 2, 3]
b = a          # b is the SAME list, not a copy
a.append(4)
b              # [1, 2, 3, 4] 🤖 b changed too

b = a.copy()   # ✓ now b is an independent (shallow) copy
import copy
b = copy.deepcopy(a) # ✓ independent even for nested lists/dicts
```

- **Trap:** `grid = [[0] * 3] * 3` makes **three references to one row** — editing `grid[0][0]` changes all rows. Use `[[0]*3 for _ in range(3)]`.
- **Rule:** assignment never copies. `=` binds another name to the same object.

□ *Bridge:* a variable is a sticky label on an object, not a container holding a value.

10 — `type()` VS `isinstance()` (Session 2 / 9)

```
isinstance(x, int)          # ✓ Pythonic; respects inheritance
isinstance(x, (int, float)) # ✓ "is x any kind of number?"
type(x) is int              # exact type only; rarely what you want
type(x) == int              # works but use `is` for type identity
```

- **Trap:** `isinstance(True, int)` is `True` (bool is a subtype of int). If you must exclude bools, check `type(x) is int` OR `isinstance(x, int)` and not `isinstance(x, bool)`.

11 — Integer vs float division (Session 1 / 2)

```
7 / 2      # 3.5    true division ALWAYS returns a float
7 // 2     # 3      floor division (rounds toward -∞)
-7 // 2    # -4     ← floors, doesn't truncate toward zero!
7 % 2      # 1      modulo (remainder) — use % 2 to test even/odd
7 / 0      # ❌ ZeroDivisionError
```

- **Trap:** `/` gives a float even when the result is whole (`4 / 2` is `2.0`). Use `//` for integer results.

12 — Strings are immutable; some methods *return*, don't mutate (Session 1 / 9)

```
s = " Hello "
s.strip()    # "Hello" ← returns a NEW string
s           # " Hello " ← s is UNCHANGED
s = s.strip() # ✓ reassign to keep the result
```

- **Trap:** `s.strip()` / `s.replace(...)` / `s.upper()` do nothing to `s` unless you reassign.
- Contrast: list methods like `.append()` / `.sort()` mutate **in place** and return `None`: `python nums = [3, 1, 2] nums = nums.sort() # ✗ now nums is None! .sort() returns None nums.sort() # ✓ sorts in place; use sorted(nums) to get a new list`

13 — `range` excludes the stop; off-by-one (Session 4)

```
list(range(1, 5))      # [1, 2, 3, 4] ← 5 is NOT included
list(range(5))         # [0, 1, 2, 3, 4]
list(range(0, 10, 2))  # [0, 2, 4, 6, 8]
```

- **Rule:** `range(a, b)` is the half-open interval `[a, b)`.

14 — Modifying a list while iterating it (Session 4)

```
xs = [1, 2, 3, 4]
for x in xs:
    if x % 2 == 0:
        xs.remove(x)    # ✗ skips elements / unpredictable
# ✓ iterate a copy, or build a new list:
xs = [x for x in xs if x % 2 != 0]
```

15 — Files: `"w"` overwrites; cursor exhausts (Session 8)

```
open("data.csv", "w")    # ! TRUNCATES the file to empty immediately
open("data.csv", "a")    # append
open("data.csv", "r")    # read (default)

with open("data.csv") as f:
    rows = f.readlines()  # read once
    again = f.readlines() # [] – cursor is at end of file now
```

- Always prefer `with open(...)` as `f`: — it closes the file even if the code crashes.
- With the `csv` module on Windows, open with `newline=""` to avoid blank rows.

16 — Regex: `.` matches *anything*; use raw strings (Session 9)

```
import re
re.search(r"\d+", "id 42")    # use r"..." so \d isn't a Python escape
re.search(".", "a.b")         # the "." matches "a", not just the dot!
re.search(r"\.", "a.b")      # use \. for a literal dot
```

- **Rule:** always write patterns as raw strings `r"..."`. Escape `.` `^` `$` `*` `+` `?` `(` `)` `[` `]` `{` `}` `|` `\`.

Quick "predict-then-run" self-test (cover the answers)

```
print(True + True)           # 2
print(3 == 3.0)              # True
print(0.1 + 0.2 == 0.3)      # False
print(5 == "5")              # False
print([1,2] == (1,2))        # False
print(bool("0"))             # True
print(7 // 2, -7 // 2)        # 3 -4
x = [1]; y = x; x.append(2); print(y)  # [1, 2]
```

If you can explain *why* for all eight, you've got the fundamentals most beginners miss.

Python Quick Reference — Syntax You'll Forget

The "how do I write that again?" sheet. Grouped by session.

I/O & types (S1)

```
name = input("Name: ")      # ALWAYS returns a str
age  = int(input("Age: "))   # convert immediately
print("Hi", name, age)       # space-separated
print(f"Hi {name}, age {age}") # f-string (preferred)
print(f"{score:.2f}")        # 2 decimals
print(f"{count:,}")          # thousands separator: 1,234
print("no newline", end="")  # suppress the trailing \n
int("42"), float("3.14"), str(42), bool(0) # casts
type(x)                       # what type is x?
```

f-string formatting mini-table

Spec	Example	Output
<code>:.2f</code>	<code>f"{3.14159:.2f}"</code>	<code>3.14</code>
<code>:,</code>	<code>f"{1234567:,}"</code>	<code>1,234,567</code>
<code>:>8</code>	<code>f"{'hi':>8}"</code>	<code>hi</code> (right-align)
<code>:<8</code>	<code>f"{'hi':<8}"</code>	<code>hi</code> (left-align)
<code>^8</code>	<code>f"{'hi':^8}"</code>	<code>hi</code> (center)
<code>:.1%</code>	<code>f"{0.873:.1%}"</code>	<code>87.3%</code>

Conditionals (S3)

```
if score >= 90:
    grade = "A"
elif score >= 80:
    grade = "B"
else:
    grade = "F"

90 <= score <= 100          # chained comparison
grade = "pass" if score >= 60 else "fail"  # ternary

match command:              # Python 3.10+
    case "start": ...
    case "stop" | "halt": ... # multiple values
    case _: ...              # default
```

Loops (S4)

```
for i in range(5): ...      # 0..4
for x in items: ...         # each element
for i, x in enumerate(items): ... # index + element
for a, b in zip(names, scores): ... # walk two lists together
while condition: ...
while True:                 # validation loop
    x = input("...")
    if valid(x):
        break
# break / continue control flow
```

Sequences & slicing (S5)

```
xs = [1, 2, 3]; xs.append(4); xs[0]; xs[-1]  # last
xs[1:3]      # [2, 3] (start inclusive, stop exclusive)
xs[:2]       # first two
xs[::-1]     # reversed
t = (1, 2)   # tuple (immutable)
d = {"name": "Ana", "gpa": 3.9} # dict
d["name"]; d.get("missing", 0)  # access; safe access w/ default
d.keys(); d.values(); d.items()
s = {1, 2, 2, 3}                # set -> {1, 2, 3} (unique)
```

Comprehensions (S5/S9)

```
[x*2 for x in xs]           # list
[x for x in xs if x > 0]     # with filter
{name: 0 for name in names} # dict
{x % 3 for x in xs}         # set
(x*x for x in xs)           # generator (lazy)
```

Sorting (S5)

```
sorted(xs)                                # new sorted list
sorted(xs, reverse=True)
sorted(students, key=lambda s: s["gpa"])    # by a field
sorted(students, key=lambda s: s["gpa"], reverse=True)
xs.sort()                                  # in place (returns None!)
```

Functions (S6)

```
def avg(nums: list[float]) -> float:
    """Return the mean of nums."""    # docstring
    return sum(nums) / len(nums)

def greet(name, greeting="Hello"): ...  # default arg (NOT mutable!)
def total(*args): ...                  # any number of positionals -> tuple
def config(**kwargs): ...              # any number of keywords -> dict
func(*my_list)                         # unpack list into args
func(**my_dict)                        # unpack dict into kwargs
```

Exceptions (S7)

```
try:
    n = int(value)
except ValueError:
    n = None
except (KeyError, IndexError) as e:
    print(e)
else:
    print("ok")          # ran only if no exception
finally:
    print("always")      # cleanup, always runs

raise ValueError("score must be 1-5")
assert n > 0, "n must be positive"
```

Files & CSV (S8)

```
with open("f.txt") as f:          # read, auto-closes
    text = f.read()
with open("f.txt", "w") as f:     # write (OVERWRITES)
    f.write("line\n")

import csv
with open("data.csv", newline="") as f:
    for row in csv.DictReader(f):  # row is a dict keyed by header
        print(row["name"])

with open("out.csv", "w", newline="") as f:
    w = csv.DictWriter(f, fieldnames=["name", "gpa"])
    w.writeheader()
    w.writerow({"name": "Ana", "gpa": 3.9})
```

Handy stdlib (S8)

```
import statistics; statistics.mean(xs); statistics.median(xs); statistics.stdev(xs)
import random; random.choice(xs); random.randint(1, 6); random.shuffle(xs)
from datetime import date; date.today(); date(2026, 6, 26)
from pathlib import Path; Path("data.csv").exists()
import json; json.dumps(obj, indent=2); json.loads(text)
```

Regex (S9)

```
import re
re.search(r"pattern", text)      # first match anywhere (or None)
re.fullmatch(r"\d{5}", zip)     # whole string must match
re.sub(r"\s+", " ", text)       # collapse whitespace
m = re.search(r"(\w+)@(\w+)", s)
m.group(1)                       # first capture group
```

Token	Means
.	any char (except newline)
\d \w \s	digit / word-char / whitespace
+ * ?	1+, 0+, 0-or-1
{m,n}	between m and n
^ \$	start / end
[...] [^...]	set / not-in-set
(...)	capture group
a b	a or b

Classes (S9)

```
class Student:
    def __init__(self, name, gpa):
        self.name = name
        self._gpa = gpa
    def __str__(self):
        return f"{self.name} ({self._gpa})"
    @property
    def gpa(self):
        return self._gpa
    @gpa.setter
    def gpa(self, value):
        if not 0 <= value <= 4:
            raise ValueError("gpa out of range")
        self._gpa = value

s = Student("Ana", 3.9)
print(s)           # uses __str__
```

The program skeleton (*every script*)

```
def main():
    ...

def helper():
    ...

if __name__ == "__main__":
    main()
```

Glossary — Plain-Language Definitions

For a learner who knows research but not code. Each term: what it is, in one breath.

- **Argument** — a value you hand to a function when you call it. `int("5")` — "5" is the argument.
- **Parameter** — the named slot in a function definition that receives an argument. `def f(x):` — `x` is the parameter.
- **Variable** — a *name* (label) pointing at an object. Not a box; a sticky note.
- **Object** — any value Python stores: a number, string, list, function, even a type. "Everything is an object."
- **Type** — the kind of an object: `int`, `float`, `str`, `bool`, `list`, `dict`, `set`, `tuple`, `NoneType`.
- **Dynamic typing** — variables aren't locked to one type; the *object* has a type, the *name* doesn't.
- **int / float** — whole number / decimal number. `/` always makes a float; `//` floors to int.
- **str** — text. Immutable (can't change in place). Written in quotes.
- **bool** — `True` / `False`. Technically a kind of `int` (`True == 1`).
- **None** — "no value." Returned by functions that don't `return` anything. Test with `is None`.
- **f-string** — `f"...{expr}..."` — a string with `{}` placeholders filled by Python.
- **Equality (`==`)** — "same value?" The comparison you usually want.
- **Identity (`is`)** — "literally the same object in memory?" Use only for `None` / `True` / `False`.
- **Truthy / Falsy** — any value used in an `if` counts as true unless it's `0`, `""`, `[]`, `{}`, `set()`, or `None`.
- **Mutable / Immutable** — can it be changed in place? `list` / `dict` / `set` yes; `int` / `str` / `tuple` no.
- **Aliasing** — two names pointing at the *same* mutable object; changing one shows in the other.
- **Slicing** — taking a sub-sequence: `xs[start:stop:step]`, stop excluded.
- **Comprehension** — a one-line way to build a list/dict/set from a loop: `[x*2 for x in xs]`.
- **Function** — a reusable named block of code; takes inputs, optionally `return`s an output.
- **return vs print** — `return` hands a value back to the *caller*; `print` just shows text to the screen.
- **Scope** — where a name is visible. Python looks Local → Enclosing → Global → Built-in (LEGB).
- ***args / **kwargs** — collect "any number of" positional args (as a tuple) / keyword args (as a dict).
- **Type hint** — optional annotation of expected types (`def f(x: int) -> str:`). Documentation, **not** enforced at runtime.
- **Exception** — an error raised at runtime (e.g., `ValueError`). Handle with `try / except`.
- **raise** — deliberately throw an exception. **EAFP** — "easier to ask forgiveness than permission": try it, catch failure.
- **assert** — a developer sanity-check that errors if false. Not for validating user input (can be turned off).
- **Module** — a `.py` file you can `import`. **Library / package** — a bundle of modules (install via `pip`).
- **Standard library** — modules that ship with Python (`csv`, `statistics`, `random`, `re`, `json`, `datetime`, `pathlib`).
- **Context manager (`with`)** — a block that sets up and tears down a resource automatically (e.g., closes a file).
- **CSV** — comma-separated values; a plain-text table. `csv.DictReader` turns each row into a dict.

- **Class / Object (OOP)** — a blueprint (`class`) and the things built from it (instances). Bundles data + behavior.
- `self` — inside a class, "this particular instance."
- `__init__` — the setup method run when you create an instance. **Dunder** = "double underscore" method.
- `@property` — makes a method look like an attribute; lets you validate on get/set.
- **Generator / `yield`** — produces values one at a time, lazily; saves memory on big data. Can be iterated once.
- `lambda` — a tiny anonymous function, often used as a `key=` for sorting.
- **Regex (regular expression)** — a pattern language for matching/extracting text (`re` module).
- **Traceback** — the error report Python prints when something fails. **Read the last line first.**
- **REPL** — the interactive `>>>` prompt where you type one line at a time.

Per-Session Quizzes (with answer keys)

Each quiz is 3–5 questions, ~5 minutes, given at the end of the session. Every question maps to a session objective. **Teacher:** ask the student to *predict* code output before they run it — the surprise is the assessment. Answers are at the end of each session block.

Session 1 — Variables & Types

1. What type does `input("x: ")` return, always?
2. What does `"5" + "3"` evaluate to? How do you get `8`?
3. What is `int(3.9)`? What is `round(3.9)`?
4. (Practical) Write one line that prints a float `score` to 2 decimal places.

Answers: 1. `str`. 2. `"53"`; use `int("5") + int("3")`. 3. `3` (truncates) and `4`.

4. `print(f"{score:.2f}")`.
-

Session 2 — Dynamic-Typing Traps

1. `a = [1,2]; b = [1,2]` — is `a == b`? is `a is b`? Why?
2. What is `True + True`? Why?
3. Is `0.1 + 0.2 == 0.3` True or False? How should you compare them?
4. What does `5 == "5"` give? What about `5 > "5"`?
5. Is `[1,2] == (1,2)` True or False? Why?

Answers: 1. `==` True (same value), `is` False (different objects). 2. `2` (bool is a subclass of int). 3. False; use `math.isclose(0.1+0.2, 0.3)` (or `round`). 4. False; `5 > "5"` raises `TypeError` (can't order int vs str). 5. False — a list and a tuple are different types.

Session 3 — Control Flow: Conditionals & Loops

1. Rewrite `if x >= 90 and x < 100:` using a chained comparison.
2. What is `5` and `0`? What is `0` or `"hi"`? Why aren't they True / False?
3. Why is `if passed == True:` not ideal? Write the better version.
4. What does `list(range(1, 5))` produce?
5. Why can removing items from a list *while looping it* go wrong, and what's a clean fix?

Answers: 1. `if 90 <= x < 100:`. 2. `0` and `"hi"` — `and` / `or` return an operand, not a bool. 3. `passed` is already a bool; write `if passed:`. 4. `[1, 2, 3, 4]` (5 excluded).

5. Removing shifts indices so the loop skips elements; iterate a copy or build a new list (`[x for x in xs if keep(x)]`).

Session 4 — Data Structures

1. Which are mutable: list, tuple, dict, set?
2. What does `xs.sort()` return? How do you get a *new* sorted list?
3. `a = [1,2]; b = a; a.append(3)` — what is `b`? Why?
4. (Practical) One-line dict comprehension mapping each name in `names` to `0`.

Answers: 1. list, dict, set are mutable; tuple is not. 2. `None` (sorts in place); `sorted(xs)`. 3. `[1, 2, 3]` — `b` is an alias for the same list. 4. `{n: 0 for n in names}`.

Session 5 — Functions & Scope

1. What's the difference between `return` and `print`?
2. Why is `def f(x, items=[])` dangerous? What's the fix?
3. What does a function return if it has no `return` statement?
4. Are type hints enforced at runtime?

Answers: 1. `return` hands a value to the caller; `print` only displays. 2. The default list is created once and persists across calls; use `items=None` then create inside. 3. `None`.

4. No — they're documentation; `mypy` checks them optionally.
-

Session 6 — Recursion & Recursive Thinking

1. What two parts must every recursive function have?
2. What error comes from a base case that's never reached, and why doesn't Python just keep going?
3. What does this `fact` return for `fact(4)`, and why?

```
python def fact(n): if n <= 1: return 1
n * fact(n - 1)
```
4. Name one situation where a plain loop is the better choice than recursion.
5. Why is recursion a natural fit for nested data (a list of lists, or nested JSON)?

Answers: 1. A **base case** (stops) and a **recursive case** (calls itself on a smaller input, moving toward the base case). 2. `RecursionError: maximum recursion depth exceeded` — Python has no tail-call optimization, so each pending call keeps a stack frame until it hits the limit (~1000). 3. `None` — the recursive case computes `n * fact(n-1)` but never `return`s it, so the function falls off the end. 4. A flat sequence, or work deep enough to exceed the recursion limit (a loop has no frame cost). 5. The data is *defined in terms of itself* (a list may contain lists), so a function defined in terms of itself mirrors its shape and can reach every level.

Session 7 — Exceptions

1. Which exception does `int("N/A")` raise?
2. Why is a bare `except:` dangerous?
3. When should you use `raise` vs `assert`?
4. (Practical) Wrap `int(value)` so it returns `None` on failure.

Answers: 1. `ValueError`. 2. It catches everything (even Ctrl+C and your own typos) and can hide bugs. 3. `raise` to validate real/untrusted input; `assert` for developer sanity checks (can be disabled). 4. `try: return int(value) except (ValueError, TypeError): return None`.

Session 8 — Files & Data

1. What does opening a file in `"w"` mode do to existing contents?
2. Why prefer `with open(...)` over `open() / close()`?
3. After `csv.DictReader`, what type is each row?
4. CSV values read from a file are what type — and what must you do with numbers?

Answers: 1. Truncates it to empty immediately. 2. `with` auto-closes the file even if the code crashes. 3. A `dict` keyed by the header row. 4. Strings; convert with `int() / float()`.

Session 9 — Regular Expressions

1. In regex, what does `.` match? How do you match a literal dot?
2. Why write regex patterns as raw strings `r"..."`?
3. Which function do you use to (a) check the *whole* string matches, and (b) replace matches?
4. What does `re.search` return when there's no match, and what must you do before `.group()`?

Answers: 1. Any character (except newline); use `\.` for a literal dot. 2. So backslashes aren't treated as Python string escapes. 3. (a) `re.fullmatch`, (b) `re.sub`. 4. It returns `None`; check `if m:` before calling `m.group()` or you'll hit an `AttributeError`.

Session 10 — Modules, OOP & Pythonic

1. In a class, what is `self`?
2. What happens if you iterate a generator twice?
3. Why doesn't a module's `if __name__ == "__main__":` block run when you `import` it?
4. What does a `@property` setter let you do that a plain attribute can't?

Answers: 1. The current instance ("this particular object"). 2. The second pass is empty — a generator is exhausted after one iteration. 3. On import, `__name__` is the module's name, not `"__main__"`, so the block is skipped. 4. Validate (or transform) the value on every assignment, so the object can reject bad data.

Scoring guide (formative, not graded)

- **All correct, explained why:** ready for the capstone.
- **Right answer, fuzzy why:** re-do that session's traps-and-gotchas rows.
- **Wrong on Session 2 items:** revisit Session 2 before continuing — it's load-bearing.